

# Pure Organic Terrace Vegetable Manual

28 vegetables with quantities, timing, container sizes, step-by-step growing, feeding, and harvest guidance

Prepared for home terrace gardening in Hyderabad and similar Indian climates

## How to use this manual

- The quantities in this guide are approximate for home containers, not field-scale farming. Always use mature, fully decomposed compost; never use fresh cow dung or raw kitchen waste directly around roots.
- “Pure organic” here means no synthetic fertilisers, herbicides, or chemical pesticides. Use organic seed, mature compost, vermicompost, neem cake, mulch, hand-picking, physical barriers, and labelled biological products where required. Certified organic production may have stricter input rules.
- Crop times are approximate. Hyderabad has hot summers, a monsoon period, and mild winters; shift the sowing date if temperatures, rainfall, or terrace exposure differ.

## Quick organic quantities

| Item                    | Home-garden quantity                                                                                                                                                        |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Base mix per 10 L       | 4 L soil + 3 L mature compost + 2 L cocopeat + 1 L coarse sand. Add 50-100 g neem cake.                                                                                     |
| Leafy crop top-dress    | 1 handful vermicompost after the first cutting or every 20-30 days.                                                                                                         |
| Fruiting crop top-dress | 1-2 handfuls vermicompost every 20-25 days after establishment; reduce feeding near final harvest.                                                                          |
| Seed treatment          | For suitable seeds, use a labelled Trichoderma or Pseudomonas product according to its label; TNAU lists Trichoderma viride at 4 g/kg seed for several horticultural crops. |
| Compost tea             | Steep 1 cup mature compost or vermicompost in 5 L water for 24 hours, strain, and apply to already-moist soil. Use the same day.                                            |
| Botanical pest spray    | Use a labelled neem formulation or a mild neem extract in the evening; test on a few leaves first and avoid spraying open flowers or harsh midday sun.                      |

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# 1. Tomato (tamatar)

| Best sowing time                                    | Approx. germination             | First harvest                   |
|-----------------------------------------------------|---------------------------------|---------------------------------|
| Jun-Sep or Oct-Feb; avoid peak April-May heat       | 5-10 days                       | 70-100 days after transplanting |
| Seed/plant quantity                                 | Container                       | Sunlight                        |
| 4-6 seeds for nursery; transplant 1 strong seedling | 20-25 L, 30-40 cm deep; 1 plant | 6-8 hours                       |

## Organic feeding quantity

At planting: 2-3 L mature compost + 50 g neem cake. Then 2 handfuls vermicompost every 20-25 days.

## Step-by-step method

- Sow shallowly in a small tray; keep moist and provide morning sun.
- Transplant at 25-30 days when the seedling has 5-6 true leaves.
- Plant deeply, leaving the top leaves above the soil; install a stake or cage immediately.
- Water at the soil surface, not over the leaves. Remove lower leaves touching the soil.
- Harvest fruits when full-sized and coloured, or earlier for cooking.

## Organic care

Mulch the surface after seedlings establish, remove diseased leaves promptly, improve airflow, and inspect the underside of leaves twice a week. For small infestations, hand-pick insects or wash them off with a gentle water spray. Use neem-based or other permitted botanical products only according to the product label, and do not spray when bees are actively visiting flowers.

Harvest note: Harvest at the stage described above and continue picking regularly where the crop is a repeat-harvest vegetable.

## 2. Green chilli (hari mirchi)

| Best sowing time                                   | Approx. germination                       | First harvest                                    |
|----------------------------------------------------|-------------------------------------------|--------------------------------------------------|
| Jun-Feb; provide afternoon shade in extreme summer | 7-14 days                                 | 90-120 days after transplanting; pick repeatedly |
| Seed/plant quantity                                | Container                                 | Sunlight                                         |
| 8-10 seeds for nursery; transplant 1-2 plants      | 10-15 L per 1 plant, or 20 L for 2 plants | 5-7 hours                                        |

### Organic feeding quantity

At planting: 1.5-2 L compost + 40 g neem cake. Add 1 handful vermicompost every 20 days.

### Step-by-step method

- Sow in a tray 0.5 cm deep and transplant at 30-40 days.
- Harden seedlings for 3-4 days by gradually exposing them to outdoor conditions.
- Pinch only the first growing tip if a bushier plant is desired.
- Keep soil evenly moist during flowering; avoid waterlogging.
- Pick green pods regularly to encourage new flowers.

### Organic care

Mulch the surface after seedlings establish, remove diseased leaves promptly, improve airflow, and inspect the underside of leaves twice a week. For small infestations, hand-pick insects or wash them off with a gentle water spray. Use neem-based or other permitted botanical products only according to the product label, and do not spray when bees are actively visiting flowers.

Harvest note: Harvest at the stage described above and continue picking regularly where the crop is a repeat-harvest vegetable.

### 3. Capsicum (bell pepper)

| Best sowing time                          | Approx. germination          | First harvest                   |
|-------------------------------------------|------------------------------|---------------------------------|
| Sep-Feb; use shade net during hot weather | 7-14 days                    | 90-120 days after transplanting |
| Seed/plant quantity                       | Container                    | Sunlight                        |
| 6-8 seeds for nursery; transplant 1 plant | 15-20 L, 30 cm deep; 1 plant | 5-7 hours                       |

#### Organic feeding quantity

At planting: 2 L compost + 50 g neem cake. Top-dress 2 handfuls vermicompost every 20-25 days.

#### Step-by-step method

- Sow in a fine, well-drained nursery mix and transplant at 35-45 days.
- Place the pot where it gets morning sun and afternoon protection in hot months.
- Support the plant with a small stake when fruits begin forming.
- Keep moisture steady; irregular watering can cause flower and fruit problems.
- Harvest green, or leave longer for coloured fruits if the variety permits.

#### Organic care

Mulch the surface after seedlings establish, remove diseased leaves promptly, improve airflow, and inspect the underside of leaves twice a week. For small infestations, hand-pick insects or wash them off with a gentle water spray. Use neem-based or other permitted botanical products only according to the product label, and do not spray when bees are actively visiting flowers.

Harvest note: Harvest at the stage described above and continue picking regularly where the crop is a repeat-harvest vegetable.

## 4. Brinjal / eggplant (vankaya)

| Best sowing time                          | Approx. germination | First harvest                   |
|-------------------------------------------|---------------------|---------------------------------|
| Jun-Feb; protect from severe summer heat  | 5-10 days           | 80-120 days after transplanting |
| Seed/plant quantity                       | Container           | Sunlight                        |
| 6-8 seeds for nursery; transplant 1 plant | 20-25 L; 1 plant    | 6-8 hours                       |

### Organic feeding quantity

At planting: 2-3 L compost + 50-75 g neem cake. Add 2 handfuls vermicompost every 20 days.

### Step-by-step method

- Raise seedlings in a tray and transplant at 30-35 days.
- Bury the root ball at the same depth; firm soil gently around it.
- Stake the plant and remove very low leaves.
- Inspect young shoots for shoot-and-fruit borer; remove affected shoots and fruits promptly.
- Harvest tender fruits regularly before seeds become hard.

### Organic care

Mulch the surface after seedlings establish, remove diseased leaves promptly, improve airflow, and inspect the underside of leaves twice a week. For small infestations, hand-pick insects or wash them off with a gentle water spray. Use neem-based or other permitted botanical products only according to the product label, and do not spray when bees are actively visiting flowers.

Harvest note: Harvest at the stage described above and continue picking regularly where the crop is a repeat-harvest vegetable.

## 5. Okra / lady's finger (bhendi)

| Best sowing time                    | Approx. germination | First harvest           |
|-------------------------------------|---------------------|-------------------------|
| Feb-Oct; sow again after heavy rain | 4-8 days            | 45-60 days after sowing |
| Seed/plant quantity                 | Container           | Sunlight                |
| 6-8 seeds; thin to 1-2 plants       | 15-20 L; 1-2 plants | 6-8 hours               |

### Organic feeding quantity

Mix 2 L compost + 50 g neem cake into the pot. Add 1 handful vermicompost at 25-30 days.

### Step-by-step method

- Soak seeds in clean water for 6-8 hours; sow 2-3 cm deep.
- Thin seedlings after 10-15 days, leaving the strongest plant or two.
- Water deeply when the top 2-3 cm dries; keep the foliage dry.
- Harvest pods every 1-2 days while they are tender.
- Use gloves if the leaves irritate your skin.

### Organic care

Mulch the surface after seedlings establish, remove diseased leaves promptly, improve airflow, and inspect the underside of leaves twice a week. For small infestations, hand-pick insects or wash them off with a gentle water spray. Use neem-based or other permitted botanical products only according to the product label, and do not spray when bees are actively visiting flowers.

Harvest note: Harvest at the stage described above and continue picking regularly where the crop is a repeat-harvest vegetable.

## 6. Cucumber (dosakaya/keera)

| Best sowing time                        | Approx. germination              | First harvest           |
|-----------------------------------------|----------------------------------|-------------------------|
| Jan-Apr and Jul-Sep; avoid waterlogging | 3-7 days                         | 45-60 days after sowing |
| Seed/plant quantity                     | Container                        | Sunlight                |
| 3-4 seeds; thin to 1-2 plants           | 20-25 L; 1-2 plants with trellis | 6-8 hours               |

### Organic feeding quantity

Use 3 L compost + 75 g neem cake at planting. Add 2 handfuls vermicompost at flowering and again after 3 weeks.

### Step-by-step method

- Sow 2-3 cm deep directly in the final pot; do not disturb roots unnecessarily.
- Install a trellis before vines become long.
- Guide vines upward and mulch the soil surface.
- Water consistently during flowering and fruit enlargement.
- Harvest young fruits frequently; over-mature fruits reduce new production.

### Organic care

Mulch the surface after seedlings establish, remove diseased leaves promptly, improve airflow, and inspect the underside of leaves twice a week. For small infestations, hand-pick insects or wash them off with a gentle water spray. Use neem-based or other permitted botanical products only according to the product label, and do not spray when bees are actively visiting flowers.

Harvest note: Harvest at the stage described above and continue picking regularly where the crop is a repeat-harvest vegetable.

## 7. Beans (French/cluster beans)

| Best sowing time                                    | Approx. germination                         | First harvest           |
|-----------------------------------------------------|---------------------------------------------|-------------------------|
| Sep-Feb for French beans; Feb-Oct for cluster beans | 4-8 days                                    | 45-65 days after sowing |
| Seed/plant quantity                                 | Container                                   | Sunlight                |
| 6-8 seeds; thin to 2-3 plants                       | 20 L; 2-3 plants, with support for climbers | 6-8 hours               |

### Organic feeding quantity

Add 2-3 L compost + 50 g neem cake. Avoid excessive nitrogen; add 1 handful vermicompost at flowering.

### Step-by-step method

- Sow directly 2-3 cm deep after soaking seeds for 4-6 hours.
- Provide a trellis for pole types; bush types need only a short support.
- Keep the root zone moist but never continuously wet.
- Mulch after seedlings establish.
- Pick pods every 2-3 days when tender.

### Organic care

Mulch the surface after seedlings establish, remove diseased leaves promptly, improve airflow, and inspect the underside of leaves twice a week. For small infestations, hand-pick insects or wash them off with a gentle water spray. Use neem-based or other permitted botanical products only according to the product label, and do not spray when bees are actively visiting flowers.

Harvest note: Harvest at the stage described above and continue picking regularly where the crop is a repeat-harvest vegetable.

## 8. Green peas (matar)

| Best sowing time                       | Approx. germination                 | First harvest           |
|----------------------------------------|-------------------------------------|-------------------------|
| Oct-Jan in Hyderabad; cool-season crop | 5-10 days                           | 60-75 days after sowing |
| Seed/plant quantity                    | Container                           | Sunlight                |
| 10-15 seeds; thin to 5-8 plants        | 20-25 L, 25-30 cm deep; use support | 5-7 hours               |

### Organic feeding quantity

Mix 3 L compost + 50 g neem cake. Add a small handful of vermicompost at flowering; do not overfeed nitrogen.

### Step-by-step method

- Soak seeds for 6-8 hours and sow 3-4 cm deep.
- Give a trellis as soon as tendrils appear.
- Keep soil cool with mulch and water when the surface dries.
- Protect from hot afternoon sun; peas set poorly in high heat.
- Harvest pods while plump but still green.

### Organic care

Mulch the surface after seedlings establish, remove diseased leaves promptly, improve airflow, and inspect the underside of leaves twice a week. For small infestations, hand-pick insects or wash them off with a gentle water spray. Use neem-based or other permitted botanical products only according to the product label, and do not spray when bees are actively visiting flowers.

Harvest note: Harvest at the stage described above and continue picking regularly where the crop is a repeat-harvest vegetable.

## 9. Spinach (palak)

| Best sowing time                          | Approx. germination                      | First harvest                          |
|-------------------------------------------|------------------------------------------|----------------------------------------|
| Sep-Feb; use shade in hot months          | 5-10 days                                | 25-40 days; cut-and-come-again         |
| Seed/plant quantity                       | Container                                | Sunlight                               |
| 2-3 g per 30 x 30 cm tray; thin seedlings | Tray or 10-15 L pot; 10-15 cm soil depth | 3-6 hours; morning sun is best in heat |

### Organic feeding quantity

Mix 1-2 L compost into the tray. Add 1 handful vermicompost after the first cutting.

### Step-by-step method

- Sow seeds 1-1.5 cm deep in rows or scatter thinly.
- Thin crowded seedlings so air can move between plants.
- Water gently and frequently while seedlings are small.
- Harvest outer leaves first, leaving the centre growing point.
- Resow every 2-3 weeks for continuous harvest.

### Organic care

Mulch the surface after seedlings establish, remove diseased leaves promptly, improve airflow, and inspect the underside of leaves twice a week. For small infestations, hand-pick insects or wash them off with a gentle water spray. Use neem-based or other permitted botanical products only according to the product label, and do not spray when bees are actively visiting flowers.

Harvest note: Harvest at the stage described above and continue picking regularly where the crop is a repeat-harvest vegetable.

## 10. Amaranth (thotakura)

| Best sowing time                       | Approx. germination                   | First harvest |
|----------------------------------------|---------------------------------------|---------------|
| Feb-Oct; grows quickly in warm weather | 3-7 days                              | 25-40 days    |
| Seed/plant quantity                    | Container                             | Sunlight      |
| 1-2 g per 30 x 30 cm tray              | Tray or 10-15 L pot; 15 cm soil depth | 4-7 hours     |

### Organic feeding quantity

Use 1-2 L mature compost in the tray; add 1 handful vermicompost after cutting.

### Step-by-step method

- Mix tiny seeds with dry sand for even sowing and cover lightly.
- Keep the surface moist until germination.
- Thin seedlings at 10-15 days.
- Cut tender shoots at 20-30 cm height; leave the base to regrow.
- Avoid harvesting a whole tray at once if repeated growth is wanted.

### Organic care

Mulch the surface after seedlings establish, remove diseased leaves promptly, improve airflow, and inspect the underside of leaves twice a week. For small infestations, hand-pick insects or wash them off with a gentle water spray. Use neem-based or other permitted botanical products only according to the product label, and do not spray when bees are actively visiting flowers.

Harvest note: Harvest at the stage described above and continue picking regularly where the crop is a repeat-harvest vegetable.

## 11. Methi / fenugreek leaves (menthulu)

| Best sowing time                      | Approx. germination                     | First harvest |
|---------------------------------------|-----------------------------------------|---------------|
| Sep-Feb; can be grown in short cycles | 3-7 days                                | 20-30 days    |
| Seed/plant quantity                   | Container                               | Sunlight      |
| 10-15 g per 30 x 30 cm tray           | Tray or 8-10 L pot; 10-15 cm soil depth | 3-6 hours     |

### Organic feeding quantity

Add 1-1.5 L compost to the tray; no heavy feeding is usually needed.

### Step-by-step method

- Soak seeds for 4-6 hours and sow 1-2 cm deep.
- Keep the soil evenly moist until seedlings emerge.
- Thin crowded patches lightly.
- Cut leaves at 10-15 cm height, leaving the crown for another flush.
- Sow a new tray every 2-3 weeks.

### Organic care

Mulch the surface after seedlings establish, remove diseased leaves promptly, improve airflow, and inspect the underside of leaves twice a week. For small infestations, hand-pick insects or wash them off with a gentle water spray. Use neem-based or other permitted botanical products only according to the product label, and do not spray when bees are actively visiting flowers.

Harvest note: Harvest at the stage described above and continue picking regularly where the crop is a repeat-harvest vegetable.

## 12. Lettuce (leaf lettuce)

| Best sowing time                     | Approx. germination                     | First harvest                               |
|--------------------------------------|-----------------------------------------|---------------------------------------------|
| Oct-Feb; needs shade in warm weather | 4-10 days                               | 40-60 days                                  |
| Seed/plant quantity                  | Container                               | Sunlight                                    |
| 0.2-0.5 g per 30 x 30 cm tray        | Tray or 8-10 L pot; 15-20 cm soil depth | 3-5 hours; protect from harsh afternoon sun |

### Organic feeding quantity

Use 2 L mature compost in the tray. Add 1 handful vermicompost after 3 weeks.

### Step-by-step method

- Sow seeds on the surface and cover with a very thin layer of mix.
- Transplant or thin to 15-20 cm spacing when seedlings have 3-4 leaves.
- Keep the mix cool and consistently moist.
- Harvest outer leaves, or cut the head above the crown.
- Provide shade during hot spells to reduce bolting.

### Organic care

Mulch the surface after seedlings establish, remove diseased leaves promptly, improve airflow, and inspect the underside of leaves twice a week. For small infestations, hand-pick insects or wash them off with a gentle water spray. Use neem-based or other permitted botanical products only according to the product label, and do not spray when bees are actively visiting flowers.

Harvest note: Harvest at the stage described above and continue picking regularly where the crop is a repeat-harvest vegetable.

## 13. Coriander (dhania)

| Best sowing time                       | Approx. germination                  | First harvest |
|----------------------------------------|--------------------------------------|---------------|
| Sep-Feb; partial shade in warm periods | 7-14 days                            | 25-40 days    |
| Seed/plant quantity                    | Container                            | Sunlight      |
| 10-15 g per 30 x 30 cm tray            | Tray or 8-10 L pot; 15 cm soil depth | 3-6 hours     |

### Organic feeding quantity

Mix 1-1.5 L compost into the tray; add a small handful of vermicompost after cutting.

### Step-by-step method

- Crush seeds gently into two halves without pulverising them; soak 6-8 hours.
- Sow 1 cm deep and keep the surface moist.
- Thin crowded seedlings or harvest them as microgreens.
- Cut stems 5-7 cm above the soil so regrowth is possible.
- Make repeat sowings every 2-3 weeks.

### Organic care

Mulch the surface after seedlings establish, remove diseased leaves promptly, improve airflow, and inspect the underside of leaves twice a week. For small infestations, hand-pick insects or wash them off with a gentle water spray. Use neem-based or other permitted botanical products only according to the product label, and do not spray when bees are actively visiting flowers.

Harvest note: Harvest at the stage described above and continue picking regularly where the crop is a repeat-harvest vegetable.

## 14. Mint (pudina)

| Best sowing time                                | Approx. germination                                | First harvest                   |
|-------------------------------------------------|----------------------------------------------------|---------------------------------|
| Year-round; best in mild weather                | Roots in 10-20 days from cuttings                  | 30-45 days after establishment  |
| Seed/plant quantity                             | Container                                          | Sunlight                        |
| Use 4-6 healthy cuttings or one rooted division | 10-15 L wide pot; keep separate because it spreads | 3-5 hours; bright morning light |

### Organic feeding quantity

Mix 2 L compost into the pot. Add 1 handful vermicompost every 30 days.

### Step-by-step method

- Take 10-15 cm non-flowering cuttings and remove lower leaves.
- Root cuttings in water or directly in moist organic potting mix.
- Plant several rooted cuttings around the pot edge.
- Keep evenly moist and trim regularly to encourage bushiness.
- Remove flower heads if the goal is leaf production.

### Organic care

Mulch the surface after seedlings establish, remove diseased leaves promptly, improve airflow, and inspect the underside of leaves twice a week. For small infestations, hand-pick insects or wash them off with a gentle water spray. Use neem-based or other permitted botanical products only according to the product label, and do not spray when bees are actively visiting flowers.

Harvest note: Harvest at the stage described above and continue picking regularly where the crop is a repeat-harvest vegetable.

## 15. Radish (mullangi)

| Best sowing time          | Approx. germination                           | First harvest |
|---------------------------|-----------------------------------------------|---------------|
| Sep-Feb; quick crop       | 3-7 days                                      | 25-45 days    |
| Seed/plant quantity       | Container                                     | Sunlight      |
| 2-3 g per 30 x 30 cm tray | 20-25 cm deep tray or pot;<br>thin to 8-10 cm | 4-6 hours     |

### Organic feeding quantity

Use loose mix with 2 L mature compost; avoid fresh manure and excess nitrogen.

### Step-by-step method

- Sow directly 1-1.5 cm deep; radish dislikes transplanting.
- Thin seedlings early so roots have space to enlarge.
- Keep moisture even to prevent splitting and harsh flavour.
- Pull one test root at the expected size.
- Harvest the remaining roots promptly when mature.

### Organic care

Mulch the surface after seedlings establish, remove diseased leaves promptly, improve airflow, and inspect the underside of leaves twice a week. For small infestations, hand-pick insects or wash them off with a gentle water spray. Use neem-based or other permitted botanical products only according to the product label, and do not spray when bees are actively visiting flowers.

Harvest note: Harvest at the stage described above and continue picking regularly where the crop is a repeat-harvest vegetable.

## 16. Carrot (gajar)

| Best sowing time            | Approx. germination                 | First harvest |
|-----------------------------|-------------------------------------|---------------|
| Oct-Jan; needs cool soil    | 7-14 days                           | 70-90 days    |
| Seed/plant quantity         | Container                           | Sunlight      |
| 0.5-1 g per 30 x 30 cm tray | At least 30 cm deep; thin to 5-8 cm | 5-7 hours     |

### Organic feeding quantity

Use sifted, stone-free mix with 2-3 L compost; avoid fresh manure and high nitrogen.

### Step-by-step method

- Sow very shallowly and cover with fine compost or cocopeat.
- Keep the surface moist until seedlings emerge; do not let it crust.
- Thin carefully, leaving one plant at each spacing.
- Do not add stones or clods that can deform roots.
- Harvest when shoulders reach the expected size.

### Organic care

Mulch the surface after seedlings establish, remove diseased leaves promptly, improve airflow, and inspect the underside of leaves twice a week. For small infestations, hand-pick insects or wash them off with a gentle water spray. Use neem-based or other permitted botanical products only according to the product label, and do not spray when bees are actively visiting flowers.

Harvest note: Harvest at the stage described above and continue picking regularly where the crop is a repeat-harvest vegetable.

## 17. Beetroot (beet)

| Best sowing time                    | Approx. germination               | First harvest |
|-------------------------------------|-----------------------------------|---------------|
| Oct-Feb                             | 5-10 days                         | 60-90 days    |
| Seed/plant quantity                 | Container                         | Sunlight      |
| 8-12 seed clusters; thin to 8-10 cm | 25-30 cm deep; 4-6 plants in 20 L | 5-7 hours     |

### Organic feeding quantity

Mix 2-3 L compost + 30 g neem cake into the container. Add 1 handful vermicompost at 30 days.

### Step-by-step method

- Soak seed clusters for 4-6 hours and sow 1.5-2 cm deep.
- Thin seedlings; each cluster may produce more than one plant.
- Water evenly while roots enlarge.
- Harvest leaves sparingly so the root can develop.
- Pull roots when they are medium-sized and tender.

### Organic care

Mulch the surface after seedlings establish, remove diseased leaves promptly, improve airflow, and inspect the underside of leaves twice a week. For small infestations, hand-pick insects or wash them off with a gentle water spray. Use neem-based or other permitted botanical products only according to the product label, and do not spray when bees are actively visiting flowers.

Harvest note: Harvest at the stage described above and continue picking regularly where the crop is a repeat-harvest vegetable.

## 18. Spring onion (green onion)

| Best sowing time                                  | Approx. germination           | First harvest |
|---------------------------------------------------|-------------------------------|---------------|
| Sep-Feb; can be grown in small batches year-round | 7-14 days                     | 45-60 days    |
| Seed/plant quantity                               | Container                     | Sunlight      |
| 1-2 g per 30 x 30 cm tray, or 15-20 sets          | 15 cm deep tray or 8-10 L pot | 4-6 hours     |

### Organic feeding quantity

Mix 1-2 L compost into the pot; add 1 handful vermicompost after 25 days.

### Step-by-step method

- Sow seeds 1 cm deep or plant healthy onion sets.
- Thin seedlings to 3-5 cm spacing.
- Keep soil moist but not soggy.
- Harvest outer stalks first, or pull the entire plant.
- Resow a small section every few weeks.

### Organic care

Mulch the surface after seedlings establish, remove diseased leaves promptly, improve airflow, and inspect the underside of leaves twice a week. For small infestations, hand-pick insects or wash them off with a gentle water spray. Use neem-based or other permitted botanical products only according to the product label, and do not spray when bees are actively visiting flowers.

Harvest note: Harvest at the stage described above and continue picking regularly where the crop is a repeat-harvest vegetable.

## 19. Bulb onion (onion)

| Best sowing time                                              | Approx. germination               | First harvest |
|---------------------------------------------------------------|-----------------------------------|---------------|
| Oct-Jan for bulb formation; spring onion is more flexible     | 7-14 days                         | 90-120 days   |
| Seed/plant quantity                                           | Container                         | Sunlight      |
| 1-2 g per 30 x 30 cm tray; transplant 1 seedling per 10-12 cm | 20-25 cm deep; 4-6 plants in 20 L | 6-8 hours     |

### Organic feeding quantity

Use 2-3 L compost + 30-50 g neem cake. Add 1 handful vermicompost at 30 days; stop heavy feeding near maturity.

### Step-by-step method

- Raise seedlings and transplant at 35-45 days, or plant sets.
- Keep the neck area from staying constantly wet.
- Provide full sun and thin crowded plants.
- When tops bend and dry, reduce watering and allow bulbs to cure.
- Dry bulbs in shade with good airflow before storage.

### Organic care

Mulch the surface after seedlings establish, remove diseased leaves promptly, improve airflow, and inspect the underside of leaves twice a week. For small infestations, hand-pick insects or wash them off with a gentle water spray. Use neem-based or other permitted botanical products only according to the product label, and do not spray when bees are actively visiting flowers.

Harvest note: Harvest at the stage described above and continue picking regularly where the crop is a repeat-harvest vegetable.

## 20. Potato (aloo)

| Best sowing time                                               | Approx. germination                  | First harvest |
|----------------------------------------------------------------|--------------------------------------|---------------|
| Oct-Feb; avoid hot months                                      | 10-20 days                           | 75-100 days   |
| Seed/plant quantity                                            | Container                            | Sunlight      |
| 2-3 certified, disease-free seed pieces with eyes per 30 L bag | 30-40 L grow bag, 30 cm or more deep | 6-8 hours     |

### Organic feeding quantity

Use 4-5 L compost + 75 g neem cake at planting. Add 2 handfuls compost when earthing up.

### Step-by-step method

- Use small whole seed potatoes or cut larger ones; let cut surfaces dry for a day.
- Plant 8-10 cm deep with eyes facing upward.
- Add more organic mix around stems as they grow (earthing up).
- Water when the upper layer dries, avoiding soggy bags.
- Harvest after foliage yellows and dries; cure tubers in shade.

### Organic care

Mulch the surface after seedlings establish, remove diseased leaves promptly, improve airflow, and inspect the underside of leaves twice a week. For small infestations, hand-pick insects or wash them off with a gentle water spray. Use neem-based or other permitted botanical products only according to the product label, and do not spray when bees are actively visiting flowers.

Harvest note: Harvest at the stage described above and continue picking regularly where the crop is a repeat-harvest vegetable.

## 21. Sweet potato (sweet potato)

| Best sowing time                | Approx. germination                  | First harvest |
|---------------------------------|--------------------------------------|---------------|
| Feb-Sep; needs warm weather     | Roots in 7-14 days from slips        | 90-120 days   |
| Seed/plant quantity             | Container                            | Sunlight      |
| 5-8 healthy vine cuttings/slips | 40-50 L wide grow bag;<br>2-3 plants | 6-8 hours     |

### Organic feeding quantity

Mix 4-5 L compost + 50 g neem cake into the bag. Avoid excess nitrogen, which makes leaves but fewer tubers.

### Step-by-step method

- Plant 20-25 cm vine slips with two or three nodes under the soil.
- Allow vines to spread or guide them around the terrace safely.
- Keep soil moderately moist during establishment, then water deeply as needed.
- Do not disturb developing tubers.
- Harvest when leaves begin yellowing and the crop is mature.

### Organic care

Mulch the surface after seedlings establish, remove diseased leaves promptly, improve airflow, and inspect the underside of leaves twice a week. For small infestations, hand-pick insects or wash them off with a gentle water spray. Use neem-based or other permitted botanical products only according to the product label, and do not spray when bees are actively visiting flowers.

Harvest note: Harvest at the stage described above and continue picking regularly where the crop is a repeat-harvest vegetable.

## 22. Bottle gourd (sorakaya/lauki)

| Best sowing time                               | Approx. germination                  | First harvest |
|------------------------------------------------|--------------------------------------|---------------|
| Feb-Apr and Jun-Sep; protect from waterlogging | 5-10 days                            | 60-90 days    |
| Seed/plant quantity                            | Container                            | Sunlight      |
| 3-4 seeds; retain 1 strong plant               | 50-60 L; 1 plant with strong trellis | 6-8 hours     |

### Organic feeding quantity

Add 5 L compost + 100 g neem cake at planting. Add 2 handfuls vermicompost every 20-25 days.

### Step-by-step method

- Soak seeds for 6-8 hours and sow 2-3 cm deep.
- Thin to one vigorous plant and build a strong overhead support.
- Guide the vine and prune only damaged or overcrowded growth.
- Water deeply during flowering and fruit development.
- Harvest tender gourds regularly before the skin hardens.

### Organic care

Mulch the surface after seedlings establish, remove diseased leaves promptly, improve airflow, and inspect the underside of leaves twice a week. For small infestations, hand-pick insects or wash them off with a gentle water spray. Use neem-based or other permitted botanical products only according to the product label, and do not spray when bees are actively visiting flowers.

Harvest note: Harvest at the stage described above and continue picking regularly where the crop is a repeat-harvest vegetable.

## 23. Ridge gourd (beerakaya)

| Best sowing time                 | Approx. germination           | First harvest |
|----------------------------------|-------------------------------|---------------|
| Feb-Apr and Jun-Sep              | 5-10 days                     | 60-90 days    |
| Seed/plant quantity              | Container                     | Sunlight      |
| 3-4 seeds; retain 1 strong plant | 40-50 L; 1 plant with trellis | 6-8 hours     |

### Organic feeding quantity

Use 4-5 L compost + 75-100 g neem cake. Add 2 handfuls vermicompost at flowering and fruiting.

### Step-by-step method

- Soak and sow seeds 2-3 cm deep directly in the final container.
- Provide a sturdy trellis before vines run.
- Mulch around the plant but keep mulch away from the stem.
- Hand-pollinate in the morning if flowers appear without fruit set.
- Harvest tender ridged fruits frequently.

### Organic care

Mulch the surface after seedlings establish, remove diseased leaves promptly, improve airflow, and inspect the underside of leaves twice a week. For small infestations, hand-pick insects or wash them off with a gentle water spray. Use neem-based or other permitted botanical products only according to the product label, and do not spray when bees are actively visiting flowers.

Harvest note: Harvest at the stage described above and continue picking regularly where the crop is a repeat-harvest vegetable.

## 24. Bitter gourd (karakaya)

| Best sowing time                 | Approx. germination           | First harvest |
|----------------------------------|-------------------------------|---------------|
| Feb-Apr and Jun-Sep              | 5-10 days                     | 55-75 days    |
| Seed/plant quantity              | Container                     | Sunlight      |
| 3-4 seeds; retain 1 strong plant | 40-50 L; 1 plant with trellis | 6-8 hours     |

### Organic feeding quantity

Mix 5 L compost + 100 g neem cake at planting. Add 2 handfuls vermicompost every 20 days.

### Step-by-step method

- Soak seeds overnight; nick the seed coat lightly only if germination is slow.
- Sow 2-3 cm deep and thin to one plant.
- Train the vine on a strong trellis with good airflow.
- Encourage pollination by transferring pollen from a male flower to a female flower in the morning.
- Harvest green, tender fruits before they turn orange.

### Organic care

Mulch the surface after seedlings establish, remove diseased leaves promptly, improve airflow, and inspect the underside of leaves twice a week. For small infestations, hand-pick insects or wash them off with a gentle water spray. Use neem-based or other permitted botanical products only according to the product label, and do not spray when bees are actively visiting flowers.

Harvest note: Harvest at the stage described above and continue picking regularly where the crop is a repeat-harvest vegetable.

## 25. Snake gourd (potlakaya)

| Best sowing time                 | Approx. germination                       | First harvest |
|----------------------------------|-------------------------------------------|---------------|
| Feb-Apr and Jun-Sep              | 7-14 days                                 | 70-100 days   |
| Seed/plant quantity              | Container                                 | Sunlight      |
| 3-4 seeds; retain 1 strong plant | 50-60 L; 1 plant with very strong trellis | 6-8 hours     |

### Organic feeding quantity

Add 5 L compost + 100 g neem cake. Top-dress with 2 handfuls vermicompost every 20-25 days.

### Step-by-step method

- Soak seeds overnight and sow 2-3 cm deep.
- Thin to one plant and support it on a strong pandal or trellis.
- Keep the growing area open and airy to reduce fungal disease.
- Hand-pollinate if fruit set is poor.
- Harvest fruits young and tender, before the skin becomes hard.

### Organic care

Mulch the surface after seedlings establish, remove diseased leaves promptly, improve airflow, and inspect the underside of leaves twice a week. For small infestations, hand-pick insects or wash them off with a gentle water spray. Use neem-based or other permitted botanical products only according to the product label, and do not spray when bees are actively visiting flowers.

Harvest note: Harvest at the stage described above and continue picking regularly where the crop is a repeat-harvest vegetable.

## 26. Pumpkin (gummadikaya)

| Best sowing time                       | Approx. germination                                                | First harvest |
|----------------------------------------|--------------------------------------------------------------------|---------------|
| Feb-Apr and Jun-Aug; needs large space | 5-10 days                                                          | 90-120 days   |
| Seed/plant quantity                    | Container                                                          | Sunlight      |
| 2-3 seeds; retain 1 plant              | 60 L or larger; 1 plant, preferably on a safe trellis or floor run | 6-8 hours     |

### Organic feeding quantity

Use 6-8 L compost + 100-150 g neem cake at planting. Add 2 handfuls vermicompost at flowering.

### Step-by-step method

- Sow 2-3 cm deep in the final large container.
- Allow one or two vines to run where the terrace can safely carry them.
- Place a board or dry mulch beneath developing fruits.
- Hand-pollinate early morning if needed.
- Harvest when the skin is hard and the stalk begins to dry.

### Organic care

Mulch the surface after seedlings establish, remove diseased leaves promptly, improve airflow, and inspect the underside of leaves twice a week. For small infestations, hand-pick insects or wash them off with a gentle water spray. Use neem-based or other permitted botanical products only according to the product label, and do not spray when bees are actively visiting flowers.

Harvest note: Harvest at the stage described above and continue picking regularly where the crop is a repeat-harvest vegetable.

## 27. Watermelon (tarbooz)

| Best sowing time                         | Approx. germination                        | First harvest |
|------------------------------------------|--------------------------------------------|---------------|
| Jan-Apr; avoid cool, waterlogged periods | 4-10 days                                  | 80-100 days   |
| Seed/plant quantity                      | Container                                  | Sunlight      |
| 2-3 seeds; retain 1 plant                | 60 L or larger; 1 plant with room for vine | 8 hours       |

### Organic feeding quantity

Use 6-8 L compost + 100 g neem cake. Add 2 handfuls vermicompost at flowering and fruit set.

### Step-by-step method

- Sow directly in the final container; watermelons dislike root disturbance.
- Keep one vigorous plant and guide the vine safely.
- Support each fruit on a sling or board if it hangs from a trellis.
- Water regularly until fruit reaches full size, then reduce excess watering as it ripens.
- Harvest when the tendril near the fruit dries and the underside turns creamy yellow.

### Organic care

Mulch the surface after seedlings establish, remove diseased leaves promptly, improve airflow, and inspect the underside of leaves twice a week. For small infestations, hand-pick insects or wash them off with a gentle water spray. Use neem-based or other permitted botanical products only according to the product label, and do not spray when bees are actively visiting flowers.

Harvest note: Harvest at the stage described above and continue picking regularly where the crop is a repeat-harvest vegetable.

## 28. Squash (summer squash/zucchini)

| Best sowing time                   | Approx. germination | First harvest |
|------------------------------------|---------------------|---------------|
| Sep-Feb; protect from extreme heat | 4-8 days            | 45-60 days    |
| Seed/plant quantity                | Container           | Sunlight      |
| 2-3 seeds; retain 1 plant          | 30-40 L; 1 plant    | 6-8 hours     |

### Organic feeding quantity

Add 4 L compost + 75 g neem cake. Add 2 handfuls vermicompost at flowering and after the first harvest.

### Step-by-step method

- Sow 2-3 cm deep in the final container.
- Leave one plant after seedlings establish and provide space for broad leaves.
- Water at the base and keep leaves dry overnight.
- Hand-pollinate morning flowers if fruits fail to form.
- Harvest young fruits frequently for best tenderness.

### Organic care

Mulch the surface after seedlings establish, remove diseased leaves promptly, improve airflow, and inspect the underside of leaves twice a week. For small infestations, hand-pick insects or wash them off with a gentle water spray. Use neem-based or other permitted botanical products only according to the product label, and do not spray when bees are actively visiting flowers.

Harvest note: Harvest at the stage described above and continue picking regularly where the crop is a repeat-harvest vegetable.

## Organic Troubleshooting

**Yellow leaves:** Check for overwatering, poor drainage, or lack of nitrogen. Let the top layer dry slightly and add a small amount of mature compost rather than a large dose.

**Seedlings falling at the base:** Usually excess moisture and poor airflow. Use clean containers, a well-drained mix, morning watering, and avoid crowding.

**Flowers but no fruits:** Common causes are heat, irregular watering, lack of pollinators, or excess nitrogen. Improve airflow, water consistently, and hand-pollinate cucurbits in the morning.

**Whiteflies, aphids, or mealybugs:** Remove heavily infested leaves, wash the plant, control ants, and use a mild labelled neem preparation in the evening after a patch test.

**Caterpillars and fruit borers:** Inspect daily, remove eggs/larvae by hand, discard damaged plant parts, and use physical netting where practical.

**Fungal spots:** Improve spacing and airflow, avoid wetting foliage at night, remove affected leaves, and use a labelled biological fungicide if necessary.

**Poor growth:** Check sunlight, root space, drainage, and whether the mix has compacted. Refresh the top 5 cm with compost and loosen the surface gently.

## Safety and terrace checklist

- Confirm that the roof can safely carry wet containers and large grow bags.
- Protect waterproofing with trays, stands, or a suitable root barrier; keep drains clear.
- Use clean water and containers. Keep compost bins covered and away from living areas.
- Keep botanical sprays away from children, pets, food, and water tanks; follow label precautions.
- Do not eat produce treated with any input until the product label waiting period has passed.

Technical basis: crop timing and spacing have been adapted from horticultural guidance such as the TNAU Agritech Portal for container-scale use. Organic practices draw on TNAU organic-farming guidance for mature organic manures, neem-based inputs, and biological seed treatments. Container quantities are practical approximations, not certification standards.

# Terrace Vegetable Growing Guide

A practical list for home terrace gardens in India, including Hyderabad

You can grow vegetables on a terrace in pots, grow bags, or suitable recycled containers. The most important requirements are adequate sunlight, good drainage, suitable container depth, and regular but controlled watering.

## Easy Terrace Vegetables

- Tomato — cherry or dwarf varieties are convenient for containers.
- Green chilli
- Capsicum
- Brinjal/eggplant
- Okra/lady's finger
- Cucumber
- Beans
- Green peas
- Spinach
- Amaranth/thotakura
- Methi/fenugreek leaves
- Lettuce
- Coriander
- Mint
- Radish
- Carrot
- Beetroot
- Spring onion
- Onion
- Potato
- Sweet potato

## Vegetables for Large Containers

- Bottle gourd/sorakaya
- Ridge gourd
- Bitter gourd
- Snake gourd
- Pumpkin
- Watermelon
- Squash

## Best Choices for Beginners

1. Spinach
2. Coriander
3. Radish
4. Green chilli
5. Tomato
6. Okra
7. Beans
8. Cucumber

## Container and Support Guide

| Plant group                   | Recommended setup                                              |
|-------------------------------|----------------------------------------------------------------|
| Leafy vegetables              | Shallow trays or pots; harvest leaves regularly.               |
| Tomato, chilli, brinjal, okra | Medium-to-large pots or grow bags; provide strong sunlight.    |
| Radish, carrot, beetroot      | Deep containers with loose soil so roots can develop.          |
| Cucumber and beans            | Containers with a trellis or vertical support.                 |
| Gourds, pumpkin, watermelon   | Large grow bags, strong support, and sufficient terrace space. |

## Important Terrace Gardening Tips

- Sunlight: Most fruiting vegetables need approximately 5–6 hours of sunlight daily.
- Drainage: Use pots or grow bags with drainage holes. Do not allow water to remain stagnant.
- Watering: Water when the top layer of soil begins to dry. Avoid overwatering.
- Soil: Use a light, well-draining mixture containing garden soil, compost, and cocopeat or another light organic material.
- Mulching: Add dry leaves or other suitable mulch to reduce moisture loss, especially during hot weather.
- Climbers: Use a secure trellis for cucumber, beans, gourds, and other climbing plants.
- Roof safety: Check waterproofing and avoid placing too many heavy wet containers in one area.
- Season: In Hyderabad, leafy vegetables are generally easier during cooler months; chilli, okra, brinjal, tomatoes, and gourds can be grown during warmer periods with careful watering and shade during extreme heat.

Note: Choose varieties labelled dwarf, bush, balcony, or container-friendly whenever available. Start with a few plants and expand after observing the sunlight and wind conditions on your terrace.